## GroundTruth

### *A Motion-Compensated Smart Cane Attachment for Ground-Level Hazard Sensing and Haptic Warnings*

**Theme:** Assistive Technology / Accessibility

**Project Type:** Hardware + Embedded AI/ML

**Proposed by:** [Team Name / Team Members]

**Institution:** Sister Nivedita University, Kolkata

**One-line concept:**

**GroundTruth adds an electronic sensing layer to a conventional white cane, helping identify and interpret ground-level changes before or during contact through ToF depth sensing, inertial motion compensation, contact sensing and selective TinyML, with simple haptic warnings.**

# 1. TITLE OF THE PROJECT / PRODUCT

## GroundTruth

### *A Motion-Compensated Smart Cane Attachment for Ground-Level Hazard Sensing and Haptic Warnings*

**GroundTruth** is a retrofit assistive-mobility system designed to add electronic ground-sensing capability to a conventional white cane without replacing the user's established cane technique.

The system focuses specifically on **ground-level information** such as downward drops, steps, pothole-like depressions and surface changes. It combines multi-zone Time-of-Flight (ToF) sensing with an IMU to compensate for the cane's continuous sweeping motion, while contact sensing and lightweight TinyML are used for selected surface-classification tasks.

The system communicates through **simple haptic patterns**, allowing the user to receive additional information without depending on a phone screen or continuous audio instructions.

# 2. PROBLEM STATEMENT

## Problem

**Conventional white-cane use primarily provides information through physical contact, which can limit advance information about ground-level changes such as drop-offs, steps, depressions and surface transitions before the cane tip reaches them.**

A conventional white cane remains an essential mobility tool because it provides direct tactile information about the environment. However, its information is fundamentally obtained through physical interaction with the ground. Research reviewed for this project identifies limited advance information for hazards such as drop-offs, potholes and surface transitions.

This matters because people with visual impairment can face increased mobility and fall-related challenges. The research package reviewed for GroundTruth reports large global populations affected by vision impairment and evidence of elevated fall risk among people with visual impairment.

The problem is therefore **not that conventional canes are ineffective**. Rather, there is an opportunity to provide an **additional layer of information** about the ground ahead while preserving conventional cane use.

### The design challenge

A useful electronic system must work while the cane itself is continuously moving.

A sensor attached to a sweeping cane cannot simply interpret raw distance measurements as though the sensor were stationary. GroundTruth therefore treats the cane as a **dynamic sensing platform** and uses IMU information to compensate for its orientation before

interpreting the ToF measurements. The proposed architecture specifically transforms the moving sensor measurements into a ground-referenced elevation profile.




# 3. SOLUTION PROPOSED

## GroundTruth: An Additional Sensing Layer for the White Cane

GroundTruth is proposed as a **split-mount retrofit system**.

Instead of placing all electronics at the cane tip, the heavier electronics and battery are mounted near the handle/upper shaft, while the lightweight sensing module is placed at the tip. This architecture is intended to preserve the normal handling characteristics of the cane while allowing the tip to perform ground sensing. The proposed design allocates approximately 110 g to the handle pod and approximately 38 g to the sensor tip.

### Core hardware

**Handle / upper shaft**

- ESP32-S3 microcontroller
- ICM-42688-P IMU
- LRA haptic actuator
- DRV2605L haptic driver
- 18650 Li-ion battery
- charging and power-management circuitry

**Sensorized tip**

- VL53L5CX/VL53L8CX multi-zone ToF sensor
- contact piezo/acoustic sensor
- experimental optical IR reflectance/water-sensing module
- impact-protected enclosure
- replaceable cane-tip interface

The proposed hardware BOM is approximately **₹3,935–₹4,465 as an initial component estimate**, not a guaranteed final manufacturing cost.

## How GroundTruth Works

### Layer 1 — Motion-aware ground sensing

The 8×8 ToF sensor generates a multi-zone distance pattern.

The IMU simultaneously measures the cane's orientation and motion.

GroundTruth combines the two to compensate for cane movement and estimate the geometry of the ground ahead. The proposed processing pipeline includes orientation estimation, coordinate transformation, ground-plane modelling and geometric anomaly detection.

## Layer 2 — Deterministic hazard detection

GroundTruth does **not** use AI for every problem.

For hazards whose geometry can be described directly, deterministic algorithms are preferred:

- downward drop
- ascending step
- pothole/depression

For example, a downward change across multiple ToF zones can be evaluated against the estimated ground plane.

This makes the system simpler and more explainable.

## Layer 3 — Contact-based surface interpretation

When the cane contacts or slides over a surface, the piezo/acoustic sensor captures mechanical vibration.

This information can potentially help distinguish surface patterns such as:

- smooth walkway
- loose gravel
- tactile paving

The research and hardware plan therefore restrict TinyML primarily to **surface/texture classification**, rather than unnecessarily applying AI to basic geometric calculations.

## Layer 4 — Confidence-aware haptic output

The system combines sensor information and applies temporal filtering/confidence logic before producing a warning.

Possible MVP haptic vocabulary:

- **single pulse:** possible ground hazard
- **repeated pulses:** persistent/approaching hazard
- **distinct texture pulse:** possible unusual surface

The exact patterns will be validated experimentally rather than assumed to be universally understood.

## Critical design principle

**GroundTruth informs; the user decides.**

It does not automatically steer the user.

It does not replace the white cane.

It does not guarantee detection.

It provides an additional information channel that can support the user's existing mobility technique.



# 4. IMPLEMENTATION PLAN

## Target Beneficiaries

The primary intended beneficiaries are:

- people who use conventional white canes;
- people with blindness or significant visual impairment;
- users navigating unfamiliar pedestrian environments;
- users who may benefit from additional information about ground-level changes.

GroundTruth is designed as a **retrofit rather than a completely new cane**, allowing the underlying conventional cane and established mobility technique to remain central.

## Development Plan

### Phase 1 — Ground-profile sensing

Build the minimum system:

- ESP32-S3
- IMU
- VL53L5CX
- one haptic actuator

First validate whether the system can maintain a stable ground estimate while the cane is swept naturally.

The research plan identifies this ground-plane recovery test as the most important early experiment.

### Phase 2 — Geometric hazard detection

Implement deterministic detection for:

- downward drops

- ascending steps
- controlled depressions

Test on a repeatable physical course.

## Phase 3 — Contact sensing

Add the piezo/acoustic sensor.

Collect controlled data from:

- concrete
- tile
- tactile paving
- gravel
- other selected surfaces.

## Phase 4 — TinyML classification

Train a lightweight model using extracted acoustic/IMU/ToF features.

The proposed architecture uses a compact model suitable for ESP32-S3 edge inference.

## Phase 5 — Experimental water sensing

The optical water-sensing mechanism will be treated as an **experimental feature**, not a guaranteed capability.

Testing will compare:

- dry concrete
- wet concrete
- shallow standing water
- reflective surfaces
- different illumination conditions.

If reliable discrimination cannot be demonstrated, the feature will not be presented as a validated capability.

### Phase 6 — Integrated prototype

Combine:

- sensing
- motion compensation
- hazard detection
- surface classification
- confidence handling
- haptic feedback.

### Phase 7 — Validation

The proposed research plan includes a campus test course covering paved walkway, tactile paving, stairs, simulated pothole, controlled water and gravel transition.

Evaluation should report:

- detection rate
- false-positive rate
- false-negative cases
- precision
- recall
- F1-score for classification
- performance by surface
- performance by lighting condition
- performance across different users.

The existing plan proposes 500 events per terrain class across five users with leave-one-user-out validation. This should be described as **prototype validation**, not proof of universal performance.

## Expected Deliverables

1. Working GroundTruth hardware prototype.
2. Retrofit cane-tip sensing module.

3. Handle-mounted processing and haptic module.
4. Ground-profile sensing algorithm.
5. Deterministic drop/step detection.
6. Prototype terrain-classification model.
7. Haptic warning interface.
8. Controlled validation dataset.
9. Performance and safety test report.
10. Demonstration course for physical evaluation.




# 5. SAFETY AND EFFICIENCY ANALYSIS

## Safety

GroundTruth is designed as a **supplementary sensing system**, not a replacement for conventional white-cane technique.

The conventional cane remains the primary physical mobility tool.

## Fail-safe principles

**1. No warning does not mean no hazard.**

Sensor failure, environmental conditions or classification uncertainty may result in missed information.

**2. Uncertain sensing should not be converted into false certainty.**

ToF confidence and signal-quality information should be monitored. Missing or invalid returns should not automatically be interpreted as “clear ground.” The hardware research specifically identifies sunlight, reflectivity and water as important ToF failure conditions.

**3. Direct sunlight is a known engineering limitation.**

Outdoor sunlight can reduce ToF performance, so GroundTruth must be tested under:

- indoor lighting
- shade
- direct sunlight
- dark surfaces
- reflective surfaces
- wet surfaces.

**4. Mechanical safety**

The tip should be:

- lightweight
- impact protected
- securely attached
- compatible with a replaceable conventional cane tip
- tested against repeated cane strikes.

The research identifies mechanical durability and user safety as two of the project's highest-risk areas.

**5. Haptic simplicity**

Too many vibration patterns may increase cognitive load.

Therefore, the MVP will use a small number of distinguishable patterns and validate recognition experimentally.

## Efficiency

GroundTruth does not claim to make walking universally “safer” or “faster.”

Its intended efficiency improvement is **information efficiency**.

Instead of relying exclusively on physical contact to discover certain ground changes, the system attempts to provide additional information from the ground ahead.

Potential benefits include:

- reduced need for repeated exploratory probing in some situations;
- earlier awareness of selected ground changes;
- no requirement for a smartphone during normal operation;
- local edge processing;
- haptic information without visual displays;
- preservation of conventional cane use.

### Safety hierarchy

**CONVENTIONAL CANE → PRIMARY MOBILITY TOOL**

**GROUNDTRUTH → ADDITIONAL SENSOR**

**USER → FINAL DECISION-MAKER**




# 6. SCALABILITY AND FUTURE DEVELOPMENT

GroundTruth is designed around a modular architecture so that sensing and processing capabilities can evolve without requiring a completely new cane.

## Scalability

### 1. Retrofit architecture

The system is designed as an attachment rather than requiring users to replace an entire white cane.

This could potentially simplify future adaptation to different cane designs, subject to mechanical compatibility testing.

## 2. Modular sensing

The sensing platform can be expanded or simplified depending on the intended use case.

Possible future modules include:

- improved depth sensing;
- additional inertial sensing;
- improved surface sensing;
- more robust environmental sensing;
- improved haptic interfaces.

## 3. Edge AI

GroundTruth is designed to process its primary functions locally on the embedded device.

This reduces dependence on:

- cloud connectivity;
- smartphones;
- continuous internet access.

## 4. Dataset expansion

Future development can expand the training and validation dataset across:

- different users;
- different walking styles;
- different cane lengths;
- different surfaces;
- weather conditions;
- indoor/outdoor environments.

## 5. Future adaptive models

Future versions could investigate personalized calibration so that the system adapts to:

- cane swing characteristics;
- user walking cadence;

- strike force;
- preferred haptic intensity.

## 6. Broader applications

The underlying sensing architecture could potentially be adapted for:

- accessibility infrastructure assessment;
- robotic ground inspection;
- mobility-assistance devices;
- terrain monitoring;
- educational embedded-AI platforms.

These are **future opportunities**, not capabilities claimed for the Stage 1 prototype.

## Cost scalability

The current component estimate is approximately ₹3,935–₹4,465. This should be presented as a **prototype-level BOM estimate**, rather than a guaranteed mass-production price.



# 7. INNOVATION ADVANTAGE

## What makes GroundTruth different?

GroundTruth does **not** claim that detecting ground hazards with electronics has never been attempted.

The competitive analysis identified existing work including **ClearWay Mobility**, which already addresses ground hazards such as potholes, drops, steps and uneven surfaces. It also identified prior patents and other electronic mobility systems.

Therefore, the innovation claim is deliberately narrower and more defensible.

## Core Innovation

### Motion-Compensated Ground Intelligence

GroundTruth combines four layers:

#### 1. Look-ahead ground profiling

An 8×8 ToF array provides spatial rather than single-point distance information.

#### 2. Cane-motion compensation

An IMU estimates cane orientation so that the moving sensor can be interpreted relative to a ground reference.

This is important because the cane is continuously sweeping rather than remaining stationary. The proposed architecture explicitly uses coordinate transformation and a sweep-phase state machine to address this problem.

#### 3. Multimodal ground verification

ToF provides geometry while contact sensing provides information about mechanical surface response.

This creates a layered sensing approach rather than relying on one sensor.

### 4. Selective TinyML

AI is used only where it is technically justified.

**Geometry → deterministic mathematics**

**Surface patterns → TinyML**

This avoids the common weakness of adding AI simply for the sake of calling a product "AI-powered."

## Defensible Innovation Statement

**GroundTruth proposes a retrofit, ground-focused sensing architecture that combines IMU-compensated multi-zone ToF ground profiling with contact-based surface sensing and selective edge TinyML, converting the resulting information into simple haptic warnings while preserving conventional white-cane use.**

### Competitive differentiation

| Existing approach | GroundTruth distinction |
|---|---|
| Conventional white cane | Adds electronic ground information |
| Single-distance sensing | Uses multi-zone spatial depth |
| Stationary sensor assumptions | Compensates for cane motion |
| AI for everything | Uses deterministic geometry + selective TinyML |
| Camera-heavy systems | Camera-free sensing architecture |
| Smartphone-dependent systems | Designed for local edge processing |
| Full smart cane replacement | Retrofit attachment concept |
| Ground detection only | Ground geometry + surface interpretation + haptic information |

The competitive research also found that WeWalk Smart Cane 2 explicitly excludes ground-level drop-off detection, while ClearWay demonstrates that ground-hazard detection itself is already an existing direction. GroundTruth's differentiation therefore rests on the **specific combination of motion compensation, multi-zone ground profiling, multimodal verification and selective TinyML**, rather than a broad “first smart cane” claim.

# 8. VISUAL IMPACT & USER JOURNEY

## BEFORE → AFTER

### Before: Conventional Cane

A user approaches an unfamiliar pathway.

The cane provides physical contact information through normal sweeping/probing. Ground changes become clearer primarily through interaction with the surface.

The user may need to slow down, probe repeatedly or interpret limited information from physical contact.

### After: GroundTruth-Assisted Cane

The user continues using the conventional cane.

GroundTruth adds another information layer:

**SENSE → UNDERSTAND → WARN → RESPOND**

The system:

1. senses the ground ahead;
2. measures cane motion;
3. compensates for sensor orientation;
4. identifies a possible ground anomaly;
5. classifies selected surface patterns;
6. communicates a simple haptic warning;

7. leaves the decision to the user;
8. allows the user to continue using normal cane skills.

# User Journey

## BEFORE

**1. Approaching an unfamiliar pathway**

User walks normally using a conventional cane.

**2. Ground change requires exploration**

A change in ground geometry may become clear mainly through physical interaction.

**3. More cautious probing**

The user may slow down or probe more carefully.

**4. Limited information before contact**

The cane provides limited advance information about some ground-level conditions.

## AFTER

**5. Normal walking with GroundTruth**

The user continues using the conventional cane.

**6. Sensing + motion understanding**

ToF and IMU measurements are combined while accounting for cane movement.

**7. Ground condition → haptic warning**

The system produces a simple vibration pattern when a relevant condition is detected with sufficient confidence.

**8. User responds and continues**

The user interprets the warning together with normal cane information and decides whether to slow down, change direction or probe further.

## Final User-Centered Message

**GroundTruth does not replace the cane. It adds another layer of information to it.**

The supplied storyboard already communicates this particularly well: the top half shows conventional cane exploration, while the bottom half shows the GroundTruth-assisted sequence of **sensing → understanding → haptic warning → informed user action**.

